

FUELS

SOLID FUELS $\longrightarrow$ 1) WOOD — READ — H₂ WORK
2) COAL —

BIOCHEMICAL STAGE
PEAT

CHEMICAL STAGE
METAMORPHISM

$2(C_6H_{10}O_5)_n \longrightarrow$ PEAT

SPONGY PEAT $\longrightarrow$ HARD, BRITTLE COAL.

Types of Coal.

Wood $\longrightarrow$ Peat $\longrightarrow$ BROWN COAL $\longrightarrow$ Bituminous $\longrightarrow$ Anthracite

$\longrightarrow$ % of Carbon content $\uparrow$, C.V $\uparrow$

$\longrightarrow$ % of N

S

V.M

Moisture $\downarrow$

BIOCHEMICAL STAGE — plant materials attacked by microorganism. end product is ~~peat~~ peat

CHEMICAL STAGE — ^{peat} ~~peat~~ buried under sedimentary rocks under high temperature and high pressure for millions of years. to form hard and brittle coal.

Proximate Analysis

The data varies with the procedure adopted so it is called as Proximate Analysis.
It gives the info about the practical utility of the coal.

- 1) Moisture content
- 2) Volatile Matter
- 3) Ash
- 4) Fixed Carbon.

Moisture content

1) Coal is always associated with moisture. When wet coal is exposed to air external moisture evaporates but still some moisture is retained called as Air dried moisture.

It is determined by heating a coal by 105-110°C in electric hot air oven for 1 hour.

After 1 hour taken out, cooled in DESSICATOR

weight ↑

$$\% \text{ of age of Moisture} = \frac{\text{Loss in wt}}{\text{wt of coal taken}} \times 100$$

2) Volatile Matter - Moisture free coal is heated in a covered Pt Crucible at 950 ± 20°C for 7 minutes. Crucible is cooled, dried in DESSICATOR

weight

$$\% \text{ age of Volatile Matter} = \frac{\text{Loss in weight}}{\text{weight of coal taken}} \times 100$$

3) Ash

composition - Silica, Alumina, Iron oxide, Small quantity of lime & Magnesia.

Coal contains inorganic mineral matter which are converted into ash by chemical reactions during combustion. It is determined by heating the residue left after the removal of volatile matter $700 \pm 50^\circ\text{C}$ for half an hour without covering.

Weight

$$\% \text{ age of Ash} = \left(\frac{\text{wt. of Residue left}}{\text{wt. of the coal}} \right) \times 100$$

(2+3 marks)

Extrinsic

- 1) Mineral matter like clay, gypsum, dirt which get mixed up during mining and the handling of coal remains as a residue after combustion.
- 2) consist of anhydrous CaSO_4 , CaCO_3 , Fe_2O_3 etc.
- 3) Its content is relatively high and can be removed mechanically.

Intrinsic

- 1) Mineral matter originally present in the vegetable matter from which coal is formed.
- 2) consist of oxides of Sodium, Potassium, Mg, Ca, Si etc.
- 3) It is relatively low in content and it cannot be removed by mechanical means.

4) Fixed Carbon

$$\% \text{ of Fixed Carbon} = 100 - \% (\text{Moisture} + \text{V.M} + \text{Ash})$$

(3 marks)

- 1) What is a fuel. How are fuels classified. Support.
- 2) Give an account on proximate analysis of solid fuel. What is significant of proximate analysis — 4 marks

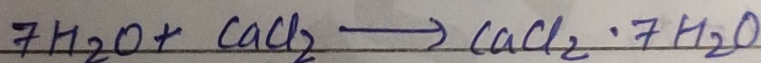
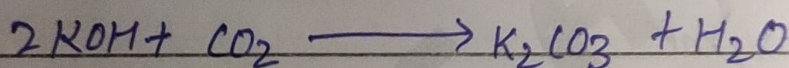
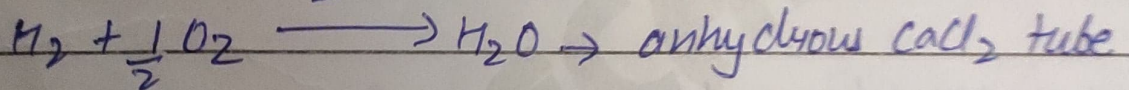
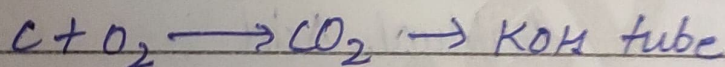
ULTIMATE ANALYSIS - To

price of coal

classification

1) Carbon & Hydrogen

Known amount of coal



wt of coal sample taken = x gram

increase in weight of $CaCl_2$ tube = y gram = H_2O

increase in weight of KOH tube = z gram = CO_2

% of carbon

44 g of CO_2 combines = 12 g of carbon

2 g of CO_2 combines = $\frac{12}{44} \times z$

% of carbon = $\frac{12}{44} \times \frac{z}{x} \times 100$

% of Hydrogen

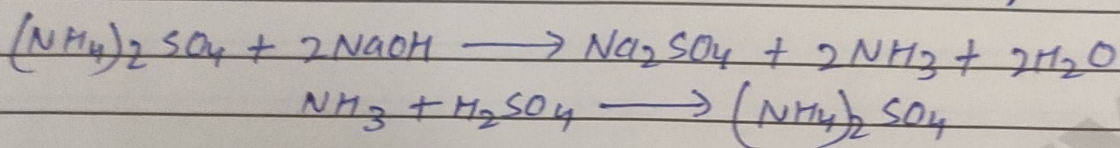
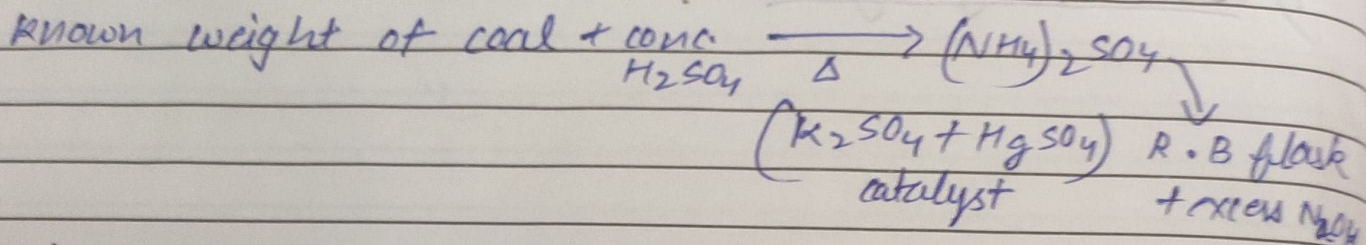
18 g of H_2O $\longrightarrow$ 2 g of Hydrogen

4 g of H_2O $\longrightarrow$ $\frac{2}{18} \times y$

$$\% \text{ of N} = \frac{1}{9} \times \frac{y}{x} \times 100$$

Nitrogen - KJEL DAHL'S METHOD

(imp)



NH_3 absorbed in acid - Rest of the acid is
 unused Acid (or)
 excess Acid (or)
 Residual Acid

Wt of coal taken = Wg

Vol. of Acid. taken = V_1 ml

Normality of Acid = N_1

(Acid) $N_1 V_1 = N_2 V_2 = (NH_3)$

$N_1 V_1 = 1 N \times V_2$ (NH_3 assumed as 1 N)

$N_1 V_1 = V_2$ — (i)

1 N Ammonia $\Rightarrow$ 1000 ml

$(1000 \text{ ml of } 1^N NH_3) = 17 \text{ g of } NH_3$
 $= 14 \text{ g of Nitrogen}$

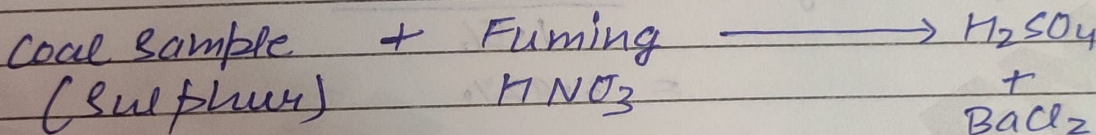
$V_2 \text{ ml of } 1 \text{ N ammonia} = \frac{14}{1000} \times V_2 \text{ g of Nitrogen}$

$$= \frac{14}{1000} \times N_1 V_1 \text{ g of Nitrogen from eq. - (1)}$$

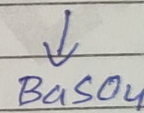
$$\% \text{ of Nitrogen} = \frac{14}{1000} \times \frac{N_1 V_1}{W} \times 100$$

$$\boxed{\% \text{ of Nitrogen} = \frac{1.4 N_1 V_1}{W}}$$

SULPHUR



Weight of coal sample taken = W g



Mass of ppt sample taken = W_1 g

$\downarrow$
Filtered (constant weight)

233g of $BaSO_4$ contains = 32g of Sulphur
 $\therefore W_1$ g of $BaSO_4 = \frac{32}{233} \times W_1$ g

$$\boxed{\% \text{ of Sulphur} = \frac{32}{233} \times \frac{W_1}{W} \times 100}$$

Ques- 1.56 g of a coal sample was KJELDAISED and ammonia gas evolved was absorbed in 50 ml of 0.1 N Sulphuric acid. After absorption, the excess or residual acid required 6.25 ml of 0.1 N NaOH for exact neutralisation. 2.6 g of a coal sample in a quantitative analysis gave 0.1755 g of $BaSO_4$. Calculate % of Nitrogen and Sulphur in the coal sample. ★

Date :

Page No:

Wt of coal taken = $w = 1.56 \text{ g.}$

Vol. of acid taken = $V_1 = 50 \text{ ml}$

$$= 50 \text{ ml} - 6.25 \text{ ml.}$$

$$= 43.75 \text{ ml.}$$

$$N_1 V_1 = N_2 \times V_2$$

$$0.01 \times 43.75 = V_2$$

$$V_2 = 4.375 \text{ ml}$$

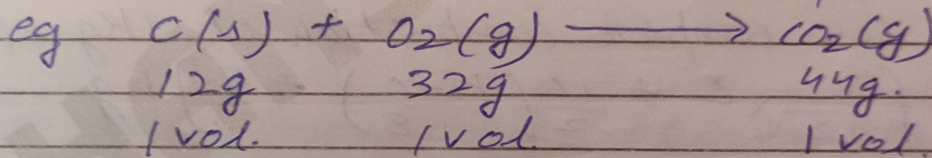
$$\% \text{ of Nitrogen} = \frac{1.4 \times 4.375}{1.56}$$

$$= 3.92\%$$

Combustion

Calculation for AIR req. for combustion

- 1) write balanced chemical equation for combustion



Amount of O_2 is calculated

- 2) combustion is actually carried out in AIR

AIR $\longrightarrow$ 21% O_2 by volume.

23% O_2 by weight

- 3) Mean molecular wt of Air = 28.94 g/mol

- 4) Composition of Solid/liq. fuel = weight

" " gaseous fuel = volume

i) Avogadro's law :- 22.4 l of gas at STP = 1 mole
 ii) $PV = nRT$

iii) Boyle's and Charles law.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

iv) Proportional volume of gases = $\frac{\text{wt of gas A}}{\text{Molar mass of gas A}}$
 $\Leftrightarrow \frac{\text{wt of gas B}}{\text{Molar mass of gas B}}$

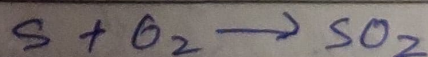
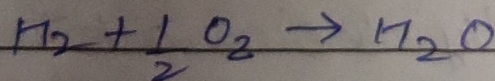
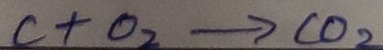
5) Min O_2 required = Theoretical O_2 - O_2 Present in fuel

% Excess AIR = $\frac{\text{Actual air used} - \text{Theoretical Air reqd}}{\text{Theoretical air}}$

b) O_2 present in fuel combines with $H_2 = \left(\frac{O}{8}\right)$

$\therefore$ Available Hydrogen = $\left(H - \frac{O}{8}\right)$

Theoretical for complete combustion of 1 kg of fuel
 $= \frac{100}{23} \left[\frac{32}{12} \times C + \frac{16}{2} \left(H - \frac{O}{8} \right) + 5 \right] \text{ kg}$



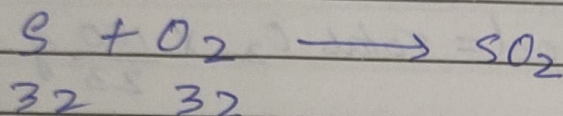
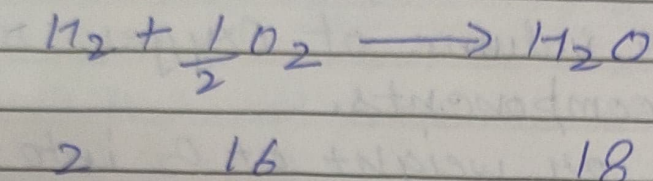
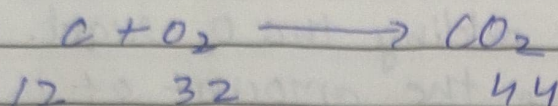
- ① Write balanced chemical equation
- ② Find total O_2 required by adding the O_2 reqd. by individual components and subtracting the amount of O_2 already present.
- ★ ③ calculate weight or volume of Oxygen required by each one of the components.
- ④ Convert the volume or weight of O_2 into air using this formula

$$\text{wt of air required} = \text{Total } O_2 \times \frac{100}{23} \text{ g}$$

$$\text{Vol. of air required} = \text{Total } O_2 \times \frac{100}{21} \text{ m}^3$$
- ⑤ Actual air supplied and excess air.
- ⑥ Convert volume into weight by avogadro law.
 let $y \text{ l}$ be the amount of air required
 $1 \text{ m}^3 = 1000 \text{ L}$
 $22.4 \text{ L} = 1 \text{ mol} = 28.94 \text{ g of air}$
 $y \text{ l} = \frac{28.94}{22.4} \times y \text{ g}$

Q. - calculate the minimum amount of air required for complete combustion of 100 kg of fuel containing carbon 80%, Hydrogen - 6%, Sulphur - 2%, Oxygen - 5% & Nitrogen by weight.

Constituent	Amount
C	$\frac{80}{100} \times 100 \text{ kg} = 80$
H	$\frac{6}{100} \times 100 = 6$
S	$\frac{2}{100} \times 100 = 2$
O	$\frac{5}{100} \times 100 = 5$

chemical eqⁿTheoretical amount of O_2

$$\text{C} - \frac{32}{12} \times 80 = 213.33$$

$$\text{H} - \frac{16}{2} \times 6 = 48$$

$$\text{S} - \frac{32}{32} \times 2 = 2$$

$$\text{Total } \underline{263.33 \text{ Kg of } \text{O}_2}$$

$$\begin{aligned} \text{Net } \text{O}_2 &= \text{Total} - \text{O present} \\ &= 263.33 - 5 \\ &= 258.33 \text{ Kg} \end{aligned}$$

$$\text{Min amt of } \text{O}_2 \text{ Air} = \text{Net } \text{O}_2 \times \frac{100}{23}$$

$$= 258.33 \times \frac{100}{23} = 1123.04 \text{ Kg gas.}$$

Following composition of Hydrogen-22%, Methane-4%, Carbon Monoxide-20%, Nitrogen-45%, Oxygen-3%.

CO_2 -6%.. i.e. 25% excess air is used. Find the weight of air actually supplied per m^3 of this gas.

Date :

Page No:

Constituent	Amount	Chemical eq ⁿ	Theoretical Air
H	$\frac{22}{100} \times 1 \text{ m}^3 = 0.22 \text{ m}^3$	$\text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{H}_2\text{O}$	$\frac{0.5}{2} \times 0.22 = 0.055$
CH ₄	$\frac{4}{100} \times 1 \text{ m}^3 = 0.04 \text{ m}^3$	$\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$	$0.04 \times 2 = 0.08$
CO	$\frac{20}{100} \times 1 \text{ m}^3 = 0.2 \text{ m}^3$	$\text{CO} + \frac{1}{2} \text{O}_2 \rightarrow \text{CO}_2$	$0.2 \times 0.5 = 0.1$
CO ₂	$\frac{6}{100} \times 1 \text{ m}^3 = 0.06 \text{ m}^3$		
O ₂	$\frac{3}{100} \times 1 \text{ m}^3 = 0.03 \text{ m}^3$		

total 0.29

$$\text{Net O}_2 = 0.29 - 0.03 = 0.26 \text{ m}^3$$

$$\text{Vol. of air} = \text{Net O}_2 \times \frac{100}{21} \text{ m}^3$$

$$= 0.26 \times \frac{100}{21} = 1.238 \text{ m}^3 \times \frac{125}{100} = 1.5475 \text{ m}^3$$

$$= 1547.5 \text{ l}$$

$$1 + 22.4 \text{ l} = 28.94 \text{ g of air}$$

$$1 \text{ l} = \frac{28.94}{22.4}$$

$$1547.5 \text{ l} = \frac{28.94}{22.4} \times 1547.5$$

$$= 1999.31 \text{ g}$$

Q- why is a good fuel for diesel engine considered a bad fuel for petroleum engine & vice-versa.
 A- How is knocking in diesel engine is different from petrol engine

Relation btw Chemical Structure & Knocking

- (1) Knocking tendency decreases with increase in compactness of molecules, = and cyclic structures.
- (2) knocking tendency in straight chain paraffins increases with increase in length of hydrocarbon.
- (3) branched chain paraffins resist knocking compared to straight chain paraffins.
- (4) Benzene and Toluene (Aromatic) have very high anti-knocking properties.

Knocking Order

PETROL ENGINE

straight chain paraffins > Branched chain paraffins > Olefins > Cyclo paraffins > Aromatic

* For diesel engine above order is reversed.

Imp Octane number and Cetane number

Ques- Why is thermal cracking better than catalytic cracking.

Ans- Autoclave

Date :

Page No:

Characteristic	Low Temp Carbonization	High Temp Carbonization
① Temp	500 - 700°C	1000 - 1200°C
② Yield	75 - 80%	65 - 75%
③ V.M	5 - 15%	1 - 3%
④ Mechanical strength	No	Yes
⑤ Hardness	Soft	Hard coke
⑥ Use	Domestic	Metallurgical
⑦ Yield of Gas	Low	High
⑧ C.V of Gas	High	Low
⑨ Aromatic %	Low	High.
⑩ straight hydrocarbons	High	Low
⑪ Smoke	Smokeless	Smoking

Ques - All caking coals are caking coals but not all caking coals are caking coals.

→ Process of carbonisation.

Regenerative Heating Principle

Date :

Page No:

- 1) Moisture and gases are driven off.
- 2) At $260-270^{\circ}\text{C}$ H_2S , H_2O , alkanes and alkenes with low molecular weight are evolved.
- 3) At 350°C decomposition of coal and evolution of gases and tarry vapours.
- 4) At 400°C caking coals are formed which are soft and plastic.
- 5) At 700°C hydrogen is evolved.
- 6) Greater than 800°C main gaseous products are evolved.
- 7) At further high temperature, it solidifies to form coke.

Read Orsat Apparatus

Recovery of product by from oven,